SUSTAINABLE ECONOMIC DEVELOPMENT IN THE CONTEXT OF EFFICIENCY

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ABSTRACT

Performance efficiency allows to improve results using the same or even reduced resources. It should become the main factor analysing sustainable development issues as well. During the research the most modern methods for increase in work efficiency, such as SPS, Six Sigma, LEAN and TOC, were overviewed, the analysis of various measures that contributed to creation of additional economic value was conducted. The research revealed that in repeated flow-based activities, the throughput is the best indicator of its efficiency.

Keywords: *efficiency of work and business, efficiency indicators, added value, throughput, investment return, sustainable development.*

INTRODUCTION

In the context of competitiveness of business development, it is particularly important to correctly evaluate efficiency of resources and outcomes. Since the results to be achieved and resources used are different, efficiency indicators, their measurement instruments and methods vary. If business aims at considerable growth of economic "harvest", the main indicator of its performance evaluation has to be economic efficiency of performance. When the business system aims at an increase in added value, its performance efficiency can be assessed applying two main indicators: throughput and investment return. Therefore it is important to evaluate the most modern theoretical and practical methods of increase in performance efficiency (LEAN system, Six Sigma, SPS and TOC).

Problem. Insufficient efficiency of business enterprises; insufficiently applied methods of added value generation.

EFFICIENCY AS A BASIS OF SUCCESS

In the scholarly literature efficiency and success of business performance is assessed through its ability to expand and compete in the market. Success of an enterprise is usually influenced by managerial skills, personal qualification of leaders, knowledge management, employees' competency and intellectual capital (Jasinavičius et al., 2007; Uzienė, 2010; Mačerinskienė & Survilaitė, 2011; Atkočiūnienė, 2013; Kalvionytė & Korsakienė, 2016; and others).

Operational efficiency is a component of what an interested individual perceives as valuable. Generally the conception of operational efficiency of an enterprise is based on the idea that an enterprise is a combination of productive assets, including human, physical and capital resources, which is used seeking to achieve the set goal. All the organised performance is based on value creation, whereas profitability is an expression of efficiency (Bartkienė, 2009, p. 94). Therefore, efficiency is seen as a multi-dimensional phenomenon, which allows for creation of added value at different levels. Generalising it can be stated that creation of added value has to be the main criteria of performance efficiency in any enterprise that targets at sustainable economic development.

MODERN VALUE GENERATION METHODS

The methodologies of process improvement such as LEAN Production, Six Sigma or the Theory of Constraints, (TOC) were created at the end of the 20th century, and their consistent application in business enterprises allowed to achieve the higher level of their performance efficiency and significantly improved economic performance.

Synchronised Production System (SPS) is based on absolute elimination of losses and a continuous seeking of production method improvement. One of the significant differences comparing it with the western LEAN Production is that implementation of SPS is a long and consistent process, which embraces involvement of top managers, production operators and employees (Cox, & Schleier, 2010). Following the results of scientific research it can be stated that only about 10 % of attempts to introduce this system of production are successful and the remaining 90 % of cases are failures (see: Lean Enterprise Institute).

The methodology of Six Sigma contributes to improvement of business processes reducing costs related to production of bad quality products. The methodology has attracted considerable attention from scholars worldwide (Breyfogle, 2003; Pyzdek, 2003; Weinstein, & Castellano, 2008; Cudney & Kanigolla, 2014; Cudney, Elrod & Stanley, 2014; Reosekar & Pohekar, 2014; Garza-Reyes, 2015; LeMahieu, Nordstrum & Cudney, 2017; and others).

Six Sigma aims at solving problems and offers valuable measurement methods, when focus is laid on process improvement and reduction of variations through specific projects (Pyzdek, 2003). Having applied this methodology, more employees get engaged in operation of an enterprise, costs for quality assurance are considerably reduced, production volumes tend to rise and time costs for completion of order decrease. Using *Six Sigma* attempts are made to strengthen attention to a client, to increase productivity, to improve financial outcomes, to reduce costs for quality assurance, to achieve and maintain quality standards.

MEASUREMENT TOC EFFICIENCY

Following the creators of TOC (Goldratt, 1990, 1994, 2002, 2003, 2015; Cox & Spencer, 1997; Dettmer, 1997; McMullen,1997; Corbett, 1998, 2005; Goldratt & Weiss, 2005; Walker & Cox, 2006; Kim, Mabin, & Davies, 2008; Cox & Schleier, 2010), the main problem of enterprises is that their top-managers rely on wrong assumptions and make endeavours to increase indicators, which are not correct in the context of the economic results in the enterprise.

Each reason, which impedes efficiency of organisation's performance, can be referred to as a constraint (Cox & Schleier, 2010). TOC is based on the assumption that *throughput* of any system is predetermined by the largest constraint of the system, which directly establishes the rate at which the system generates money. It is determined as the rate at which the system generates goal units. In business it is money received from added value. Observing changes in this indicator, heads of a company can immediately establish if their decisions increase profitability of their company or not (Corbett, 2005). On the basis of TOC, economic results of any enterprise can be revealed using the following three indicators: throughput, investment and operating expense.

E. M. Goldratt states that (Goldratt, 1990, 1994, 2002, 2003, 2015), H. W. Dettmer (1997) and T. B. McMullen (1997) the total costs of an enterprise equals the sum of costs of all the branches just as the weight of the chain is equal to the sum of the weights of its elements. They indicate compatibility of performance assessment indicators in a company. The abovementioned system of indicators can be applied in the management of an enterprise seeking an increase in the enterprise's profitability. The main formulas for throughput calculation are presented below:

$Tu = P - TVC$, (1), where: Tu – throughput for output unit; P – price per unit of output; TVC – variable costs of unit of output.

$TTp = Tu \times q$, (2), where: TTp – throughput time of production; q – quantity of output units sold through this time; an enterprise's throughput = ΣTTp.

Having calculated throughput of output in an enterprise and being aware of its operational costs and investments, the net profit (NP) and return on investment (ROI) can be connected to daily activities of the head of an enterprise. This can be illustrated by the following formula:

$NP=T-OE$, (3)

$ROI = (T-OE)/I$, (4), where: T – throughput, OE – operational expense and I – investment.

Possessing three indicators (T, I and OE), it possible to calculate the impact of decision on the enterprise's final outcome. The ideal decision would be about an increasing T and decreasing I and OE. It is important to ensure an equal increase in all the elements of the chain. In such a case when throughput of one of the element of the change goes down, throughput of all the participants will decrease, too.

Practical achievements show that TOC is compatible with and supplements other methodologies such as JIT (Just–In–Time), LEAN and Six Sigma. The study conducted by APICS (The Association for Supply Chain Management) in 2006 confirm the fact that applying Six Sigma or LEAN together with TOC synergetic effect is achieved, which is 24 times as efficient as only Six Sigma and 10 times as efficient as only LEAN. TOC opens a path for support of thinking and management, therefore it is necessary to solve problems of efficiencies in an enterprise as soon as possible, to be flexible to gain competitiveness and advantage over partners, rather than delay search for solutions until tomorrow, when a solved problem cannot bring any use (Jasinavičius et al, 2007). Thus, "using theory of constraints in project structures, their efficiency is enhanced" (Sarapinas & Sūdžius, 2009, p. 73). Further, practical application of TOC methodology in an enterprise is discussed.

PRACTICAL APPLICATION OF TOC METHODOLOGY IN AN ENTERPRISE

Most frequently heads of enterprises follow the principle of safety, seeking to "to survive" at least in the short run. Encountering problems, such companies make attempts to fight with symptoms rather than with causes of the symptoms. It is a short-term and useless tactics. Therefore, an enterprise of a retail trade network was chosen for the research.

Lost positions in the competitive market, decreasing sales volumes, insufficient added value and a fall in profitability caused a crisis in the enterprise. Therefore, it was decided to introduce the TOC methodology.

The audit in the enterprise revealed the prevailing problems of supply chain and stock management, which resulted in undesirable consequences and symptoms. The most significant stock management problems were related to consequences of inappropriate stock management: surplus or shortage.

Seeking to reduce lost sales, the enterprise used to keep large stocks because the process of stock supply used to last several weeks and suppliers were not always punctual. Information about stocks was not available on the internet, orders were generated manually and placed from thirty points of sale. Goods were not grouped into assortments, shelf space was inefficiently used: the same goods were placed on shelfs, which only performed the function of warehousing. The enterprise frequently faced depreciation of goods, their physical and moral obsolescence, goods expiration time and their writing off. Unplanned quantities of goods during sales, poor management of stock in the central warehouse and in points of sale as well as frequent inventories caused a decrease in sales volumes.

The main problems in supply change and stocks management are presented in the *current reality tree* (CRT) drawn by the authors of the research following the methodology of TOC, which will be presented during the conference.

GENERALISATION

It was established that after implementation of TOC methodology, the data of throughput observation allowed to better forecast and model alternative scenarios of activity implementation. In other worlds, all this allowed to link and harmonise various actions and to achieve better outcomes of common activity, to evaluate work of organisation in a systemic way, various activity opportunities and perspectives, to protect from threats coming from external environment.

The research results revealed that the enterprise not only improved sales results but also optimised assortment, introduced an efficient system of processes of ordering goods and implementation activities and accelerated them. All this resulted in an increase in productivity of employees and a significant decrease in order execution time. Unnecessary and non-value added activities were withdrawn. Appropriate management of processes allowed to avoid operational downtimes, surplus or shortage of output and stocks, repeated work, to improve relations with suppliers and customers. Positive changes were achieved having changed an attitude, reformed processes and withdrawn unsound habits and preconditions.

Summarising it can be stated that the achieved results allowed to increase a throughput of an enterprise creating favourable conditions for creation of added value, making attempts to link internal and external processes into a unified complex in an enterprise, targeting at achievement of common goals and eliminating contradictions among separate business processes. The process of decision making in an enterprise has become much more transparent, manageable and measurable processes have been introduced and long-term competitive advantage has been ensured.

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